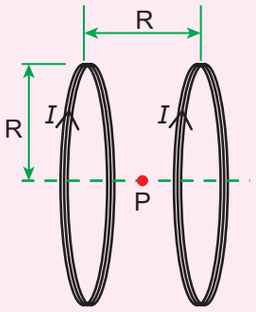
magnetic field at a point P at a distance of $R/2$ from the centre of each coil is



(a) $\frac{8N\mu_{\circ}I}{\sqrt{5}R}$ (b) $\frac{8N\mu_{\circ}I}{5^{3/2}R}$

(c) $\frac{8N\mu_{\circ}I}{5R}$ (d) $\frac{4N\mu_{\circ}I}{\sqrt{5}R}$

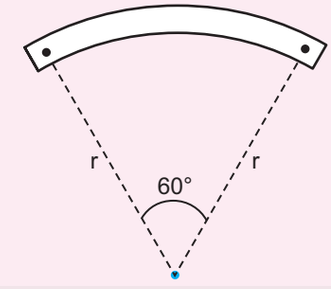
8. A wire of length l carrying a current I along the Y direction is kept in a magnetic field given by $\vec{B}=\frac{\beta}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})T$. The magnitude of Lorentz force acting on the wire is

(a) $\sqrt{\frac{2}{3}}\beta Il$ (b) $\sqrt{\frac{1}{3}}\beta Il$

(c) $\sqrt{2}\beta Il$ (d) $\sqrt{\frac{1}{2}}\beta Il$

9. A bar magnet of length l and magnetic moment p_m is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be

(NEET 2013)



(a) p_m (b) $\frac{3}{\pi}p_m$

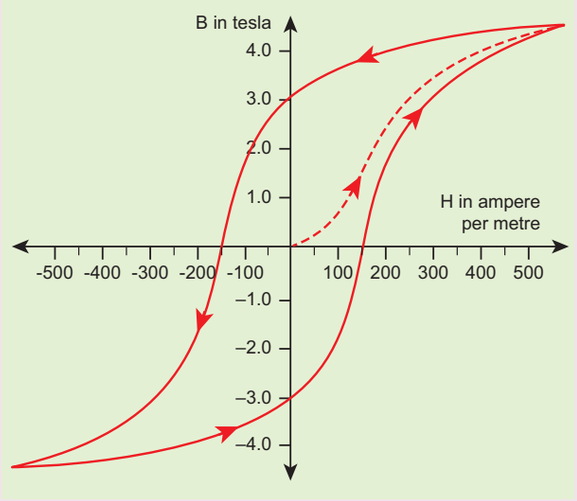
(c) $\frac{2}{\pi}p_m$ (d) $\frac{1}{2}p_m$

10. A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about its axis with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is

(a) $\frac{q}{m}$ (b) $\frac{2q}{m}$

(c) $\frac{q}{2m}$ (d) $\frac{q}{4m}$

11. The BH curve for a ferromagnetic material is shown in the figure. The material is placed inside a long solenoid which contains 1000 turns/cm. The current that should be passed in the solenonid to demagnetize the ferromagnet completely is



(a) 1.00 m A (b) 1.25 mA

(c) 1.50 mA (d) 1.75 mA

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